W11L01

30 - April - 2024

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## The class P : time complexity class.

$$P = \bigcup_{k \geq 0} \text{TIME}(n^k)$$

Polynomial time

PATH

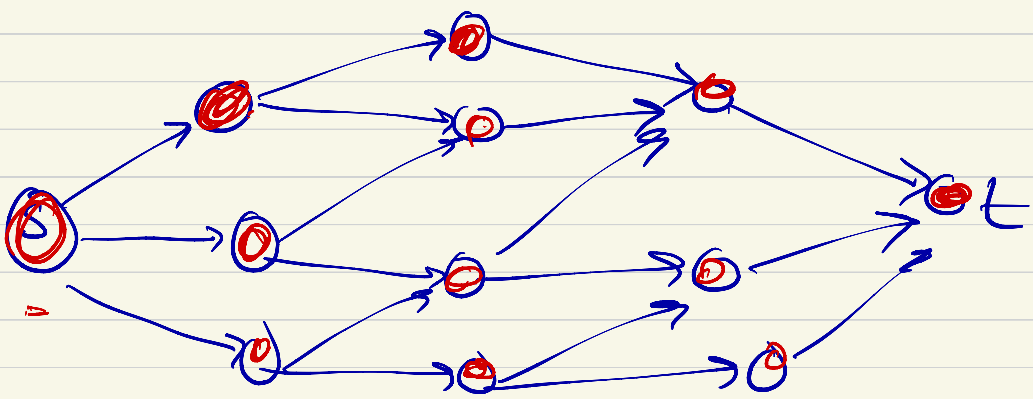
Graph : Directed graph

s, t

is there a path from s to t in G?

PATH = { <G, s, t> | G is a directed graph and there is a directed path from s to t }

PATH is decidable




PATH ∈ P

"PRIMES is in P."

$$\text{PRIMES} = \{ q \mid q \text{ is a prime no} \}$$

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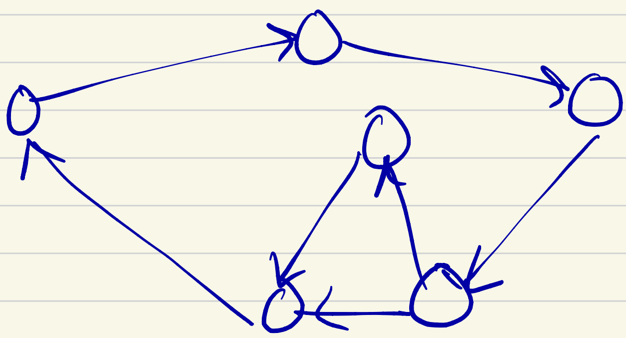
Class NP :

$$\boxed{NP = \bigcup_{k \geq 0} \text{NTIME}(n^k)}$$

$$\text{HAMILTONIAN} = \{ \langle G \rangle \mid G \text{ has a Hamiltonian path} \}$$

Suppose G is a directed graph

then a Hamiltonian of a graph is a path that goes through each vertex of G exactly once.



$HAMPATH = \{ \langle G, s, t \rangle \mid$ G is a directed graph & contains a hamiltonian path from s to t $\}$

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$PATH = \{ \langle G, s, t \rangle \mid$ G is directed graph & there exists a directed path from s to t $\}$

$\overline{PATH} = \{ \langle G, s, t \rangle \mid$ G is directed graph & there does not exist a path from s to t $\}$

$\overline{HAMPATH}$

# Alternate definition of NP.

① $$NP = \bigcup_{k \geq 0} NTIME(n^k)$$

② We say a language $L \in NP$ if there exists a verification algorithm that checks that a given solution (certificate) is a valid solution in polynomial time on a deterministic TM.